Parvez Ali

M.Sc. Artificial Intelligence & Data Science | Machine Learning & Medical Imaging Researcher

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RESEARCH INTERESTS

Artificial intelligence and machine learning for healthcare and medical imaging, with interests in deep learning, computer vision, clinical decision support, and translational AI. Particularly interested in developing robust, interpretable, and clinically useful models for complex biomedical data and bridging the gap between AI research and real-world healthcare applications.

EDUCATION

M.Sc. Artificial Intelligence and Data Science (Joint Degree) **Mar 2024 – Sep 2026**

Deggendorf Institute of Technology (THD), Germany & University of South Bohemia (USB), Czech Republic

- **Final grade:** 1.5 (THD) and 1.4 (USB), on a scale where 1.0 is the best grade
- **Master's thesis:** Predicting Anti-VEGF Injection Needs in Neovascular Age-Related Macular Degeneration (nAMD) Using Longitudinal OCT Data

B.E. Electrical Engineering (Computer Systems) **Aug 2018 – Aug 2022**

Sukkur IBA University, Sukkur, Pakistan

- **CGPA:** 3.46 / 4.00
- **Bachelor's thesis:** Deep-learning-based voice authentication for the Sindhi language. Collected a dataset of 100+ single-channel voice recordings from 12 speakers and trained an RNN-based model reaching 82% test accuracy.

RESEARCH & PROFESSIONAL EXPERIENCE

deepeye Medical GmbH, Germany **May 2025 – Present**

Software Engineer (Jul 2026 – Present)

Data Scientist Working Student (Nov 2025 – Jun 2026) | Intern (May 2025 – Nov 2025)

- Develop Python-based DICOM integration services connecting ophthalmic imaging platforms (Heidelberg HEYEX, ZEISS FORUM, Topcon Harmony/IMAGEnet) with PACS and EMR/EHR systems (TOMEDO, HEXABIT, FIDUS), including DICOM anonymization and SCP/SCU communication.
- Deploy and maintain production AI/ML infrastructure using Python, GCP, GKE, and distributed computing frameworks such as Ray.
- Worked with longitudinal retinal OCT data, developing temporal machine learning models for binary classification of treatment need from historical clinical observations.
- Develop and evaluate ML/DL models for retinal OCT imaging and clinical decision support in retinal diseases, including nAMD.
- Build OCT data pipelines covering curation, preprocessing, quality control, and dataset preparation for model training and evaluation.

Subject Specialist, Computer Science **Aug 2023 – Mar 2024**

IBA Community Colleges, Pakistan

- Taught Web Development (MERN stack) and Object-Oriented Programming (Java) to BS Computer Science students.
- Created self-paced tutorials and course material supporting different learning styles and independent study.

Artificial Intelligence Trainer **Jun 2023 – Aug 2023**

MAQ Group of Companies, Pakistan

- Designed and implemented a Dialogflow chatbot within a React Native app for health-related support and product recommendations.

Engineering Intern **Jun 2021 – Aug 2021**

Sukkur Electric Power Company, Pakistan

- Analyzed testing data for transformers and circuit breakers and built dashboards communicating system health to engineering teams.

SELECTED PROJECTS

Road Segmentation from Satellite Imagery

- Implemented a U-Net segmentation model on the Massachusetts Roads Dataset (1,000+ images).

Genetic Sequence Analysis in Bioinformatics

- Analyzed GC content, codon usage and neutrality plots across three species (Cyprinus carpio, Rattus norvegicus, Parus major) and built a Random Forest classifier to identify the genomic features most predictive of species.

TECHNICAL SKILLS

Machine & Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, CNNs, RNNs/LSTMs, time-series modelling

Medical Imaging: OCT, DICOM, medical image processing, PACS/EMR integration, pynetdicom

Deployment & Infrastructure: Flask, REST APIs, GCP, GKE, Ray

Data: SQL, HDF5, Apache Spark, Power BI

LLMs & Agents: LangChain, LangGraph, Llama, Retrieval-Augmented Generation

LANGUAGES

English (Fluent) · German (Basic) · Urdu (Native)